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(54) **SYSTEMS AND METHODS FOR REMOTE
POWER TOOL DEVICE CONTROL**

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CPC **B25F 5/00** (2013.01); **G08C 17/02**
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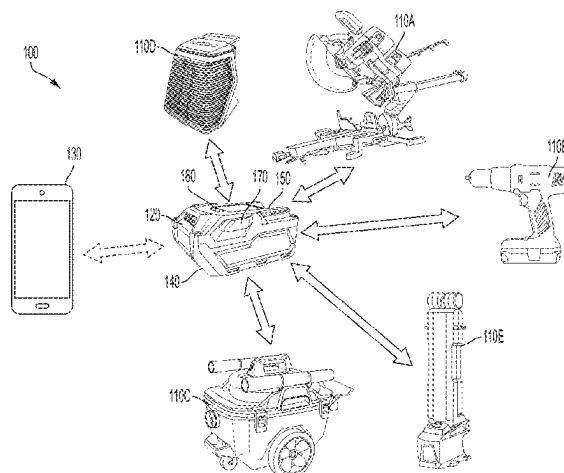
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(57) **ABSTRACT**

Systems and methods for remote power tool control are
provided. In one example, a battery pack is coupled to a
power tool device. The battery pack includes a pack trans-
ceiver and a pack electronic processor. The pack electronic
processor is coupled to the pack transceiver and is config-
ured to determine that the power tool device is remotely
controllable. The pack electronic processor is further con-
figured to receive, wirelessly via a pack transceiver of the
battery pack, a remote control command from a mobile
device, and to provide the remote control command to the
power tool device. The system further includes a tool
electronic processor of the power tool device in communi-
cation with the pack electronic processor. The tool electronic

(Continued)



processor is configured to control the power tool device to perform an action specified by the remote control command in response to receiving the remote control command.

20 Claims, 6 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/236,442, filed on Apr. 21, 2021, now Pat. No. 11,386,774, which is a continuation of application No. 16/524,970, filed on Jul. 29, 2019, now Pat. No. 11,011,053.

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H01M 10/42 (2006.01)
H02J 7/00 (2006.01)

(52) **U.S. Cl.**
 CPC . **H02J 7/00032** (2020.01); **H01M 2010/4271** (2013.01); **H01M 2010/4278** (2013.01); **H01M 2220/30** (2013.01); **H02J 7/0063** (2013.01)

(58) **Field of Classification Search**
 CPC ... H01M 2220/30; H02J 7/0063; Y02E 60/10; B25F 5/00
 USPC 340/12.5, 7.38, 6.8; 700/160, 168
 See application file for complete search history.

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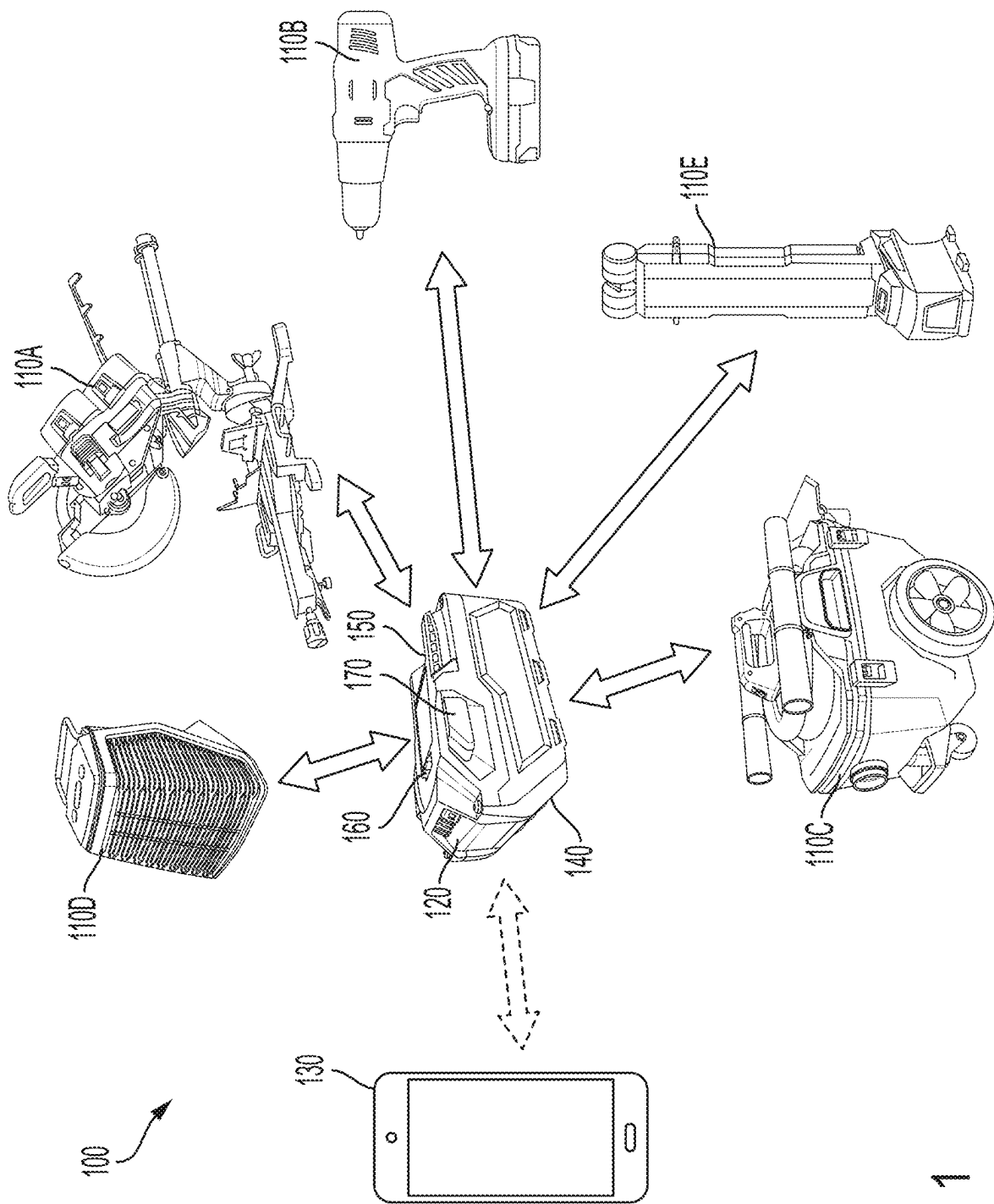


FIG. 1

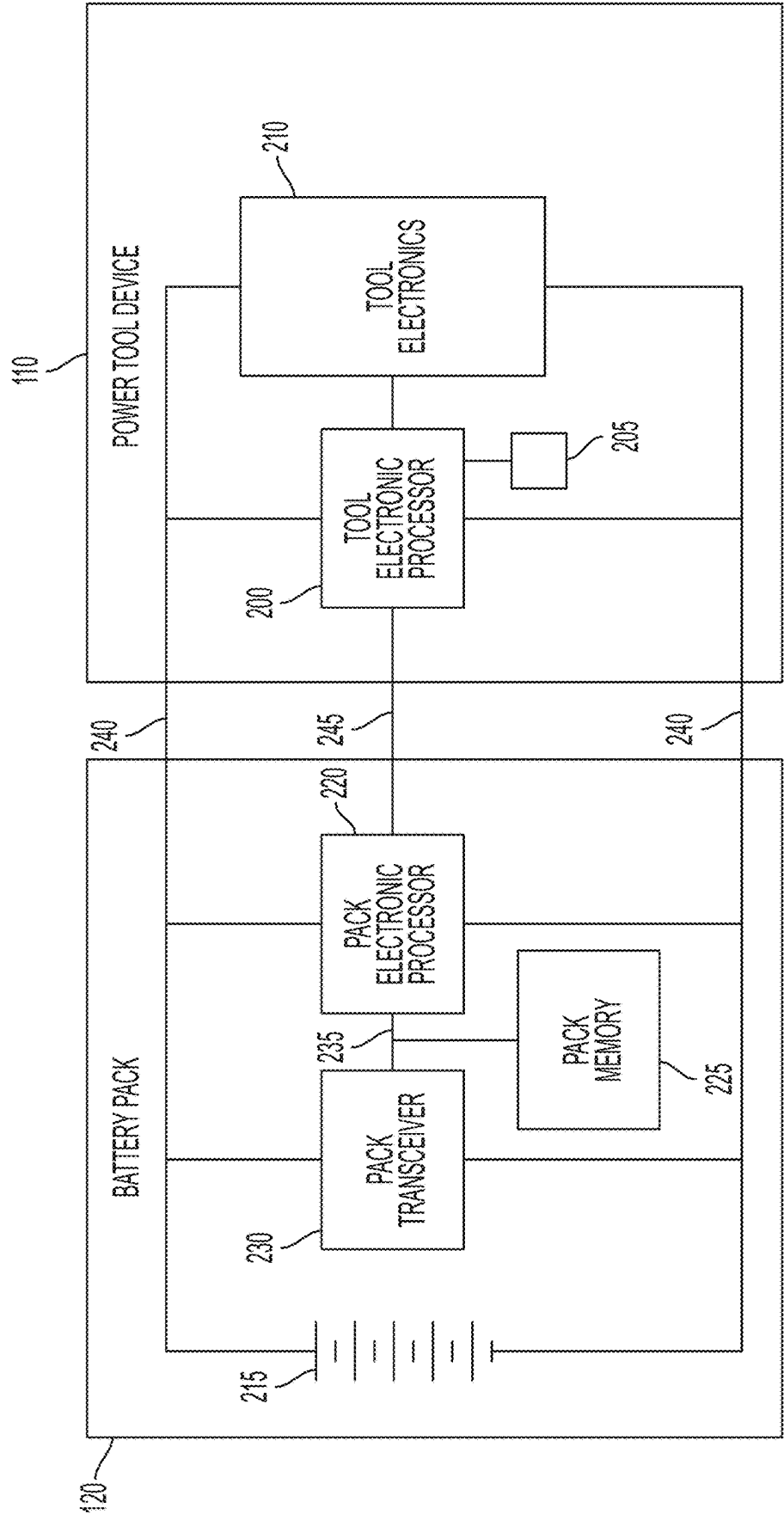


FIG. 2

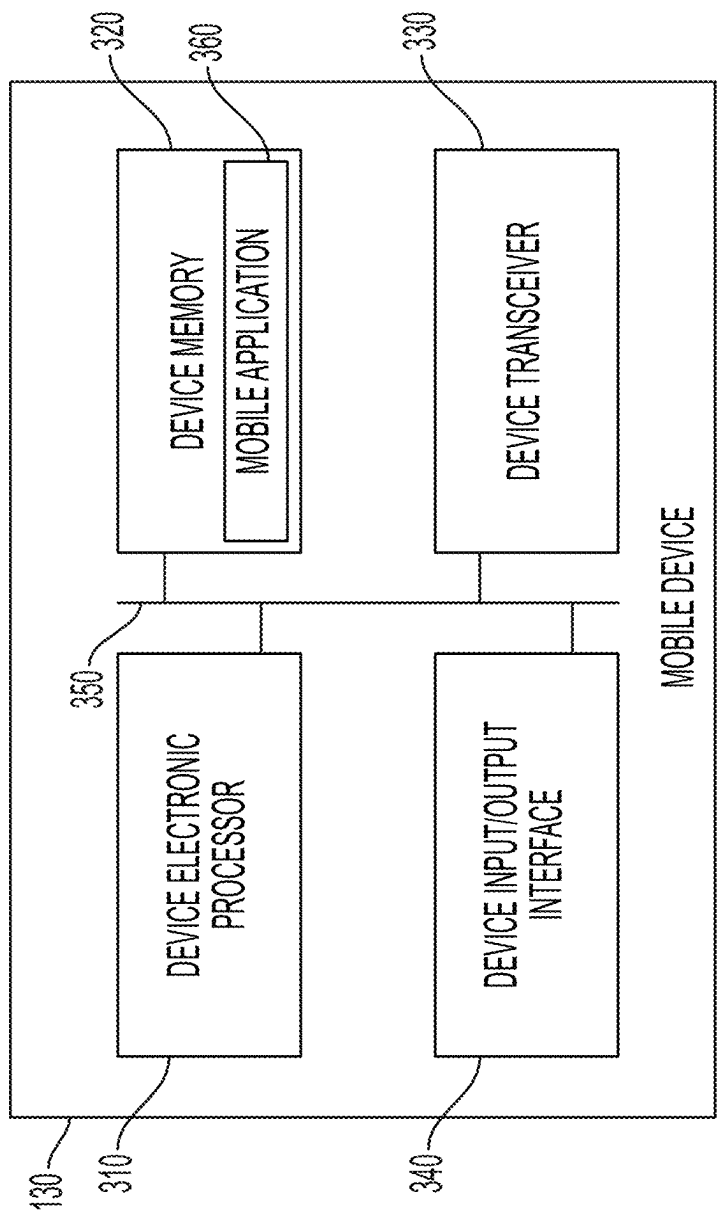


FIG. 3

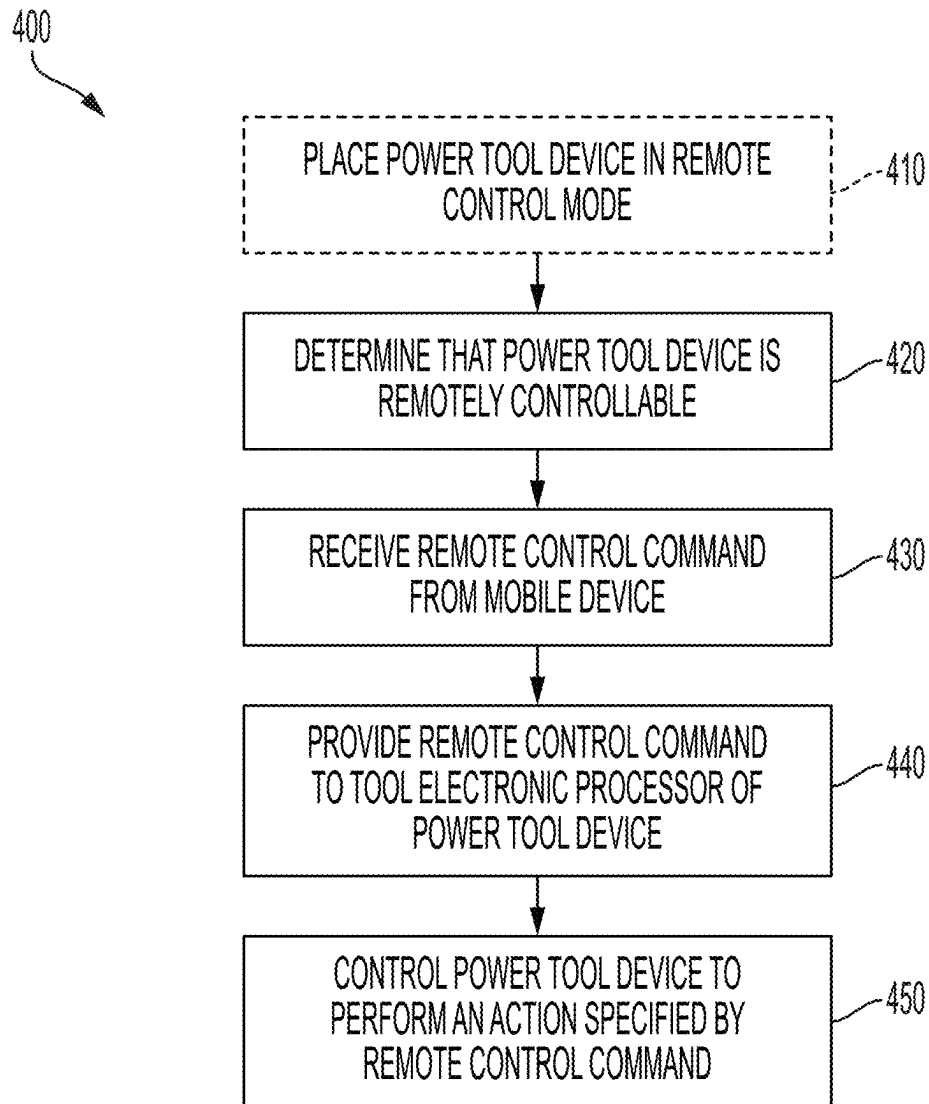


FIG. 4

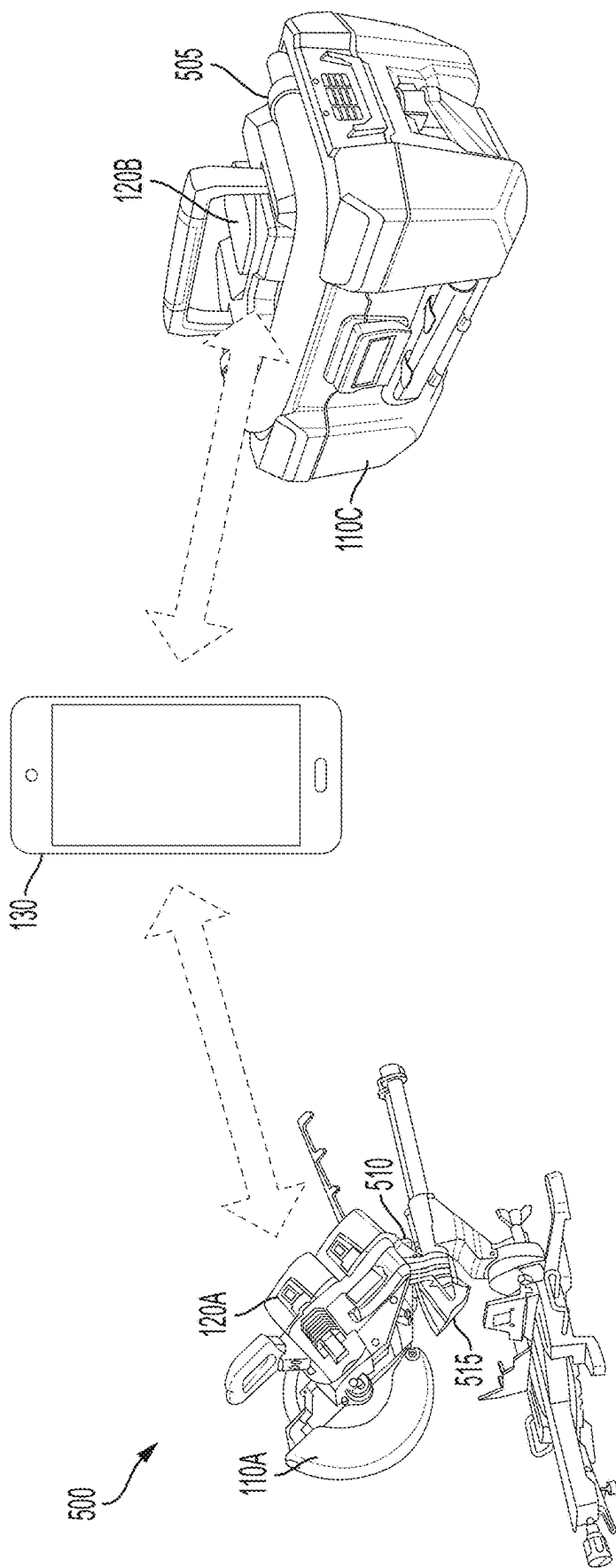


FIG. 5

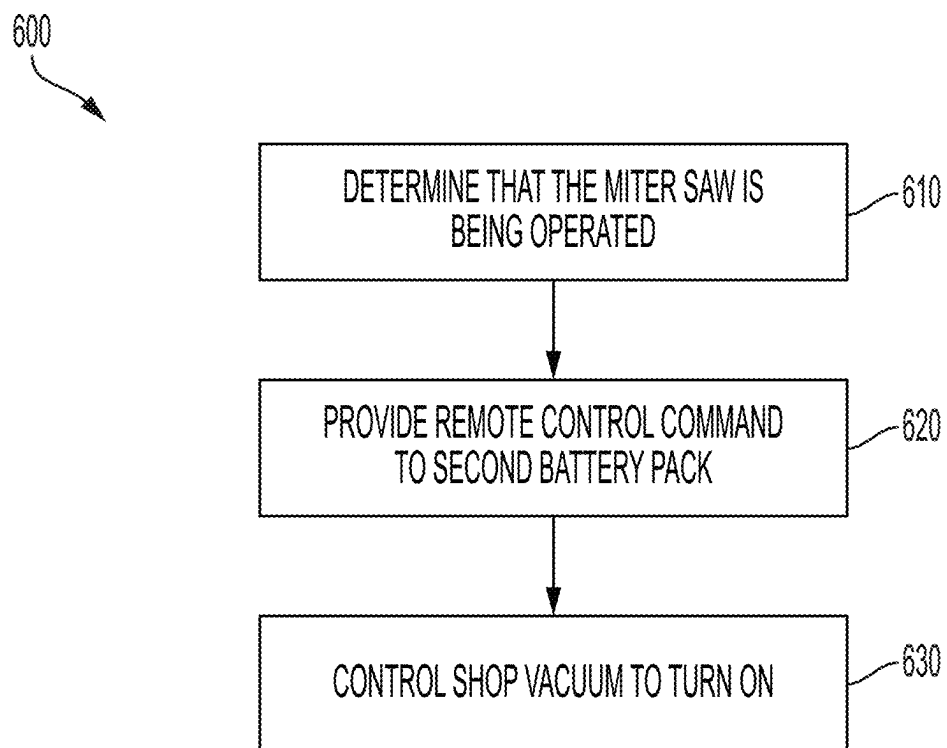


FIG. 6

SYSTEMS AND METHODS FOR REMOTE POWER TOOL DEVICE CONTROL

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/812,107, filed Jul. 12, 2022, now U.S. Pat. No. 11,890,738, which is a continuation of U.S. patent application Ser. No. 17/236,442, filed Apr. 21, 2022, now U.S. Pat. No. 11,386,774, which is a continuation of U.S. patent application Ser. No. 16/524,970, filed Jul. 29, 2019, now U.S. Pat. No. 11,011,053, which claims the benefit of U.S. Provisional Patent Application No. 62/712,473, filed on Jul. 31, 2018, the entire content of each of which is hereby incorporated by reference.

FIELD OF THE INVENTION

This application relates to controlling power tools with a mobile device through a battery pack of the power tool.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a communication system including a mobile device, a battery pack, and power tools.

FIG. 2 is a block diagram of the battery pack and the power tool of FIG. 1 in accordance with some embodiments.

FIG. 3 is a block diagram of the mobile device of FIG. 1 in accordance with some embodiments.

FIG. 4 is a flowchart of a method for remotely controlling the power tool device of FIG. 1 in accordance with some embodiments.

FIG. 5 is a system for communication between a miter saw and a shop vacuum through battery packs and the mobile device in accordance with some embodiments.

FIG. 6 is a flowchart of a method for automating dust collection during operation of the miter saw of FIG. 5 in accordance with some embodiments.

DETAILED DESCRIPTION OF THE INVENTION

Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways.

One embodiment discloses a system for remote controlling a power tool device. The system includes a battery pack coupled to a power tool device. The battery pack includes a pack memory, a pack transceiver, and a pack electronic processor. The pack electronic processor is coupled to the pack memory and the pack transceiver and is configured to determine that the power tool device is remotely controllable. The pack electronic processor is further configured to receive, wirelessly via a pack transceiver of the battery pack, a remote control command from a mobile device, and to provide the remote control command to the power tool device. The system further includes a tool electronic processor of the power tool device in communication with the pack electronic processor. The tool electronic processor is configured to control the power tool device to perform an action specified by the remote control command in response

to receiving the remote control command. In some examples, the tool electronic processor is further configured to place the power tool device in a remote control mode in response to user input.

Another embodiment provides a method for remote controlling a power tool device. The power tool device is powered by a battery pack. The method includes determining, by a pack electronic processor of the battery pack, that the power tool device is remotely controllable and receiving, wirelessly via a pack transceiver of the battery pack, a remote control command from a mobile device. The method also includes providing the remote control command, by the pack electronic processor to the tool electronic processor of the power tool device, and controlling, using the tool electronic processor, the power tool device to perform an action specified by the remote control command in response to the tool electronic processor receiving the remote control command. In some examples, the method further includes placing the power tool device in a remote control mode in response to user input.

Another embodiment provides a battery pack connectable to a power tool device and configured to facilitate remote control of the power tool device by a mobile device. The battery pack includes a plurality of cells providing operating power to the power tool device, wherein the power tool device is coupled to the battery pack and a pack transceiver. The battery pack also includes a pack electronic processor electrically coupled to the transceiver. The pack electronic processor is further configured to determine that the connected power tool device is remotely controllable and receive, wirelessly via the pack transceiver, a remote control command from the mobile device. The pack electronic processor is also configured to provide, via a communication link between the pack electronic processor and a tool electronic processor of the power tool device, the remote control command. The remote control command specifies an action to be performed by the power tool device. In some examples, the power tool device performs the function specified by the remote control command in response to receiving the remote control command.

FIG. 1 illustrates a communication system 100 including various power tool devices 110 powered by a battery pack 120. The system 100 also includes a mobile device 130 that can control the power tool devices 110 through the battery pack 120. The power tool devices 110 may include motorized power tool devices (for example, a miter saw 110A, a drill-driver 110B, a shop vacuum 110C, and the like) or non-motorized electrical devices (for example, a work radio 110D, a work light 110E, and the like). Each of the power tool devices 110A-E may be individually referred to as the power tool device 110, or collectively as the power tool devices 110. At least in some embodiments, the power tool devices 110 may be described as electrically powered devices that are configured to be coupled to and powered by a power tool battery pack (e.g., the battery pack 120) that is configured to be coupled to and power a motorized power tool (e.g., a drill, a saw, and the like).

The battery pack 120 is a power tool battery pack having a nominal voltage of, for example, 12 Volts, 18 Volts, and the like. The battery pack 120 includes a housing 140, a tool interface 150, and a latch 160 controlled by actuator 170 to selectively latch the tool interface 150 to a battery interface of the power tool 110. The mobile device 130 is a mobile communication device, for example, a smart telephone, a tablet computer, a laptop computer, a personal digital assistant, a smart wearable device (e.g., smart watch), and the like.

With reference to FIG. 2, the power tool device 110 includes a tool electronic processor 200, a tool memory 205, and tool electronics 210. The tool electronic processor 200 may be implemented as, for example, a microprocessor, a microcontroller, a field programmable gate array, an application specific integrated circuit, or the like. The tool memory 205 may be a part of the tool electronic processor 200 or may be a separate component. The tool memory 205 may include, for example, a program storage area and a data storage area. The tool memory 205 stores executable instructions that when executed by the tool electronic processor 200, cause the power tool device 110 to perform the functions described herein. For example, the tool electronic processor 200 controls the functions of the power tool device 110 and enables communication between the power tool device 110 and the battery pack 120. The tool electronics 210 may include a switch bridge and a motor (not shown) when the power tool device 110 is a motorized power tool and may include other electronics (e.g., LEDs, radio transceiver, speaker, and the like) when the power tool device 110 is a non-motorized electronic device. The tool electronics 210 are controlled by the tool electronic processor 200. For example, the tool electronic processor 200 is configured to one or more of enable the tool electronics 210, disable the tool electronics 210, and modify operating characteristics (e.g., motor power, LED brightness, radio tuning, speaker volume, and the like).

The battery pack 120 includes battery cells 215, a pack electronic processor 220, a pack memory 225, and a pack transceiver 230 within the housing 140. The pack electronic processor 220, the pack memory 225, and the pack transceiver 230 communicate over one or more control and/or data buses (for example, a communication bus 235). The battery cells 215 may be arranged in a series, parallel, or series-parallel combination. For example, the battery cells 215 include one or more series strings of five cells connected in parallel. In some embodiments, the battery cells 215 have a lithium-ion based chemistry and each provide approximately 3.6 nominal voltage. In other embodiments, the battery cells 215 have different chemistry, voltage output, or both. The battery cells 215 provide operating power to the other components of the battery pack 120. Additionally, operating power from the battery cells 215 is provided to the power tool device 110 over power terminals 240.

The pack electronic processor 220 may be implemented as, for example, a microprocessor, a microcontroller, a field programmable gate array, an application specific integrated circuit, or the like. The pack memory 225 may be a part of the pack electronic processor 220 or may be a separate component. The pack memory 225 may include, for example, a program storage area and a data storage area. The pack memory 225 stores executable instructions that when executed by the pack electronic processor 220, cause the battery pack 120 to perform the functions described herein. The pack electronic processor 220 communicates with the tool electronic processor 200 over a communication terminal 245 to exchange data and control signals. The communication terminals 245 may implement a serial communication system for example, an RS-485 link or the like to facilitate communications between the pack electronic processor 220 and the tool electronic processor 200. In some embodiments, rather than over the communication terminal 245, the pack electronic processor 220 and the tool electronic processor 200 may communicate over near-field wireless communication link, for example, a Bluetooth® connection or the like. In such embodiments, the power tool device 110 and

battery pack 120 include respective wireless transceivers to facilitate the wireless communications.

The pack transceiver 230 facilitates communication between the battery pack 120 and an external device, for example, the mobile device 130 over a wireless communication network. In some embodiments, the pack transceiver 230 includes a combined transmitter-receiver component. In other embodiments, the pack transceiver 230 includes separate transmitter and receiver components.

The power tool device 110 and the battery pack 120 may include more or fewer components and may perform functions other than those described herein.

With reference to FIG. 3, the mobile device 130 includes a device electronic processor 310, a device memory 320, a device transceiver 330, and device input/output interface 340. The device electronic processor 310, the device memory 320, the device transceiver 330, and the device input/output interface 340 communicate over one or more control and/or data buses (for example, a communication bus 350). The mobile device 130 may include more or fewer components and may perform functions other than those described herein.

The device electronic processor 310 may be implemented as, for example, a microprocessor, a microcontroller, a field programmable gate array, an application specific integrated circuit, or the like. The device memory 320 may store executable instructions that are executed by the device electronic processor 310 to carry out the functionality of the mobile device 130 described herein.

The device transceiver 330 facilitates communication between the mobile device 130 and an external device, for example, the battery pack 120 over a wireless communication network. In some embodiments, the device transceiver 330 includes a combined transmitter-receiver component. In other embodiments, the device transceiver 330 includes separate transmitter and receiver components. The device transceiver 330 is controlled by the device electronic processor 310, for example, to transmit and receive data between the mobile device 130 and the battery pack 120.

The device input/output interface 340 may include one or more input mechanisms (e.g., a keypad, a mouse, and the like), one or more output mechanisms (e.g., a display, a speaker, and the like), or a combination of the two (e.g., a touch screen, or the like).

The mobile device 130 also includes a mobile application 360, which is an application designed for a mobile operating system for use on the mobile device 130. The device memory 320 may store the mobile application 360 and the device electronic processor 310 executes the mobile application 360 to enable the mobile device 130 to carry out the functionality of the mobile application 360 described herein. The mobile application 360 may communicate with the battery pack 120 over a connection between the mobile device 130 and the battery pack 120. The mobile application 360 may include a graphical user interface in that, execution of the mobile application 360 by the device electronic processor 310 may generate a graphical user interface on a display (e.g., input/output interface 340) of the mobile device 130. The mobile device 130 may convey information to a user through display of the graphical user interface and may receive user input via the graphical user interface (i.e., the input/output interface 340).

In some embodiments, the mobile device 130 (via the device transceiver 330) and the battery pack 120 (via the pack transceiver 230) communicate over a direct wireless connection, for example, a Bluetooth® connection, a Zig-Bee® connection, or the like. In other embodiments, the

5

mobile device **130** (via the device transceiver **330**) and the battery pack **120** (via the pack transceiver **230**) communicate over an indirect wireless connection, for example, over a cellular network, over the Internet, or the like.

FIG. **4** is a flowchart illustrating an exemplary method **400** for remotely controlling the power tool device **110**. As illustrated in FIG. **4**, the method **400** includes placing the power tool device **110** in a remote control mode (at step **410**). The power tool device **110** may include a mode switch (not shown) that can be actuated by a user to select a mode of the power tool device **110**. For example, the mode switch may be between a first position for selecting the remote control mode and a second position for selecting a normal mode (i.e., deselecting the remote control mode) by the user. The tool electronic processor **200** receives position information of the mode switch and places the power tool device **110** in the selected mode. Particularly, the tool electronic processor **200** determines that the mode switch is in the first position and places the power tool device **110** in the remote control mode. When the power tool device **110** is in the remote control mode, the power tool device **110** can be remotely controlled by the mobile device **130** as described below. When the power tool device **110** is in the normal mode, the power tool device **110** ignores (e.g., discards) commands received from the mobile device **130** without executing the received commands. However, the step **410** of placing the power tool device **110** in a remote control mode is optional. For example, in some embodiments, the power tool device **110** may always be in a remote control mode such that the power tool device **110** can be remotely controlled by the mobile device **130**.

The method **400** also includes determining, by the pack electronic processor **220**, that the power tool device **110** is remotely controllable (at step **420**). The remote control feature may not be provided on every power tool device **110** configured to be coupled to and powered by the battery pack **120**. For example, the remote control feature may be provided on the work radio **110D**, the work light **110E**, and the shop vacuum **110C**, but may not be provided on the miter saw **110A** or the drill-driver **110B**. In some embodiments, the pack electronic processor **220** determines whether the power tool device **110** is remotely controllable using identification signals received from the power tool device **110**. For example, the tool electronic processor **200** communicates identification signals over the communication terminal **245** to the pack electronic processor **220**.

The identification signals may include for example, a type of the power tool (e.g., by model number), which is then used by the pack electronic processor **220** to access and retrieve from a lookup table an indication of whether the power tool device **110** is remotely controllable. The lookup table may be on the stored on the pack memory **225**, the device memory **320**, or a combination thereof. In some embodiments, the identification signals include an explicit indication of whether the power tool device **110** is remotely controllable or not remotely controllable.

In some embodiments, the battery pack **120** includes a sensor in communication with the pack electronic processor **220** that is configured to detect whether the power tool device **110** is remotely controllable. For example, the sensor of the battery pack **120** may be a Hall effect sensor configured to detect a magnetic field, and the power tool device **110** that is remotely controllable may include a magnet near its battery pack interface. Upon coupling the power tool device **110** and the battery pack **120**, the sensor provides an output to the pack electronic processor **220** indicative of the presence (or absence) of the magnet or indicative of the pole

6

orientation of the magnet, and the output is indicative of whether the power tool device **110** is remotely controllable. Accordingly, power tool devices **110** having no such magnet, or having a magnet with a pole orientation representing that the device is not remotely controllable, are determined by the pack electronic processor **220** to be not remotely controllable. Power tool devices **110** having a magnet, or having a magnet with a pole orientation representing that the device is remotely controllable, are determined by the pack electronic processor **220** to be remotely controllable.

The method **400** further includes receiving, by the pack electronic processor **220**, a remote control command from the mobile device **130** (at step **430**). The remote control command can be a command to, for example, turn the power tool device ON/OFF, activate a motor of the power tool device, switch an LED ON/OFF, adjust a radio station tuning, adjust an LED brightness, adjust a speaker volume, adjust a motor speed, and the like. The command can be selected on a graphical user interface of the mobile application **360**. The battery pack **120** may communicate the type or identification information of the power tool device **110** connected to the battery pack **120** to the mobile device **130**. The mobile device **130** may display a list of commands a user can select on the graphical user interface of the mobile application **360**. When the mobile device **130** receives a selection of the remote control command from the list of commands (e.g., based on user input received by the device input/output interface **340**), the mobile device **130** transmits the remote control command to the battery pack **120** via the device transceiver **330**. Particularly, the pack electronic processor **220** receives the remote control command wirelessly via the pack transceiver **230**.

The method also includes providing, by the pack electronic processor **220**, the remote control command to the tool electronic processor **200** of the power tool device **110** (at step **440**). The pack electronic processor **220** relays the command received from the mobile device **130** to the tool electronic processor **200**. As described above, the pack electronic processor **220** and the tool electronic processor **200** communicate over the communication terminal **245** or over a near-field communication link. The pack electronic processor **220** provides the remote control command to the tool electronic processor **200** via the communication terminal **245** or the near-field communication link. In some embodiments, the pack electronic processor **220** may provide remote control command in response to determining that the power tool device **110** is remotely controllable, that the power tool device **110** is in a remote control mode, or both.

The method **400** further includes controlling, by the tool electronic processor **200**, the power tool device **110** to perform an action specified by the remote control command (at step **450**). The tool electronic processor **200**, in response to receiving the remote control command, controls the tool electronics **210** to perform the action specified by the remote control command. For example, the tool electronic processor **200** turns the power tool device ON/OFF, activates a motor of the power tool device, switches an LED ON/OFF, adjusts a radio station tuning, adjusts an LED brightness, adjusts a speaker volume, adjusts a motor speed, and the like. In some embodiments, the power tool device **110** operates in a lower power draw mode until a remote control command is received from the battery pack **120**. In the low power draw mode, the power draw is sufficient to maintain communication with the battery pack **120** and monitor for remote control commands, but not sufficient to perform the actions specified by the remote control command. Upon receiving

the remote control command, the tool electronic processor **200** switches the power tool device **110** to the high power draw mode to perform the action specified by the remote control command.

While the steps of the method **400** are illustrated in a particular serial order, in some embodiments, one or more of the steps are executed in parallel or in a different order than illustrated. For example, one or both of steps **410** and **420** may occur in parallel with or after step **430**.

FIG. **5** illustrates one example system **500** for implementing a remote controlling of power tool devices **110**. The system **500** includes the miter saw **110A** (for example, a first power tool device) connected to a first battery pack **120A**, the shop vacuum **110C** (for example, a second power tool device) connected to a second battery pack **120B**, and the mobile device **130**. The first battery pack **120A** and the second battery pack **120B** are examples of the battery pack **120** described above. Accordingly, the description provided above with respect to the battery pack **120** similarly applies to the first battery pack **120A** and the second battery pack **120B**. The mobile device **130** wirelessly communicates with the first battery pack **120A** and the second battery pack **120B** as described above.

When the miter saw **110A** is operated on a workpiece, the resulting cut may create dust that is deposited on the work bench. Users may use the shop vacuum **110C** to clear the dust deposited by the miter saw **110A**. However, the user may have to pause the current cut to vacuum excess dust, or operate the vacuum between successive cuts to clear dust. This dust removal may result in a user taking additional time to complete a project. In some embodiments, a hose **505** of the shop vacuum **110C** is directly coupled to a dust port **510** of the miter saw **110A**. The dust port **510** includes a dust intake end **515** near the saw blade to extract dust during a cut and a dust exhaust end, opposite the dust intake end **515**, to expel extracted dust into the hose **505** coupled to the dust port. Still, users may need to manually turn on and off the shop vacuum **110** with each cut, or leave the shop vacuum **110** enabled between cuts despite a lack of dust needing extraction between cuts.

The dust collection process can be automated to be more efficient and to speed up the project by remotely controlling the shop vacuum **110C** while the miter saw **110A** is being operated. FIG. **6** is a flowchart illustrating an exemplary method **600** for automating dust collection during operation of a miter saw **110A**. As illustrated in FIG. **6**, the method **600** includes determining, by the device electronic processor **310**, that the miter saw **110A** (i.e., the first power tool device **110**) is being operated (at step **610**). For example, a pack electronic processor of the first battery pack **120A** detects a power draw when the user operates the miter saw **110A**, for example, using a current sensor electrically connected to the power terminals of the first battery pack **120A**. The pack electronic processor of the first battery pack **120A** sends a signal, via the pack transceiver, indicating that the miter saw **110A** is being operated to the mobile device **130** in response to detecting the power draw. The device electronic processor **310** of the mobile device **130** determines that the miter saw **110A** is being operated upon receiving this signal from the first battery pack **120A**.

The method **600** also includes providing, by the device electronic processor **310**, a remote control command to the second battery pack **120B** in response to the determination that the miter saw **110A** is being operated (at step **620**). For example, in response to determining that the miter saw **110A** is being operated, the device electronic processor **310** transmits the remote control command via the device transceiver

330, and the remote command is received by the second battery pack **120B**. The remote control command is a request to turn the shop vacuum **110C** (i.e., the second power tool **110**) ON (i.e., to activate a motor of the shop vacuum **110C**).

The method further includes controlling the shop vacuum to turn ON in response to the remote control command received by the second battery pack **120B** (at step **630**). For example, the pack electronic processor **220** of the second battery pack **120B** relays the remote control command to the tool electronic processor **200** of the shop vacuum **110C**. In response to the remote control command, the tool electronic processor **200** of the shop vacuum **110C** switches the shop vacuum **110** from the low power draw mode to the high power draw mode and activates the motor of the shop vacuum **110C**. Accordingly, the shop vacuum **110C** may be operated essentially simultaneously with the miter saw **110A** without any user intervention. In other words, when the miter saw **110A** is activated by the user, the shop vacuum **110C** is activated. This allows the dust collection process to be automated, which saves time for the user and provided a more efficient dust extraction.

In some embodiments, a similar technique is used to deactivate the shop vacuum **110C** in response to deactivation of the miter saw **110** by the user. For example, after step **630**, the device electronic processor **310** determines that the miter saw **110A** (i.e., the first power tool device **110**) has ceased being operated. For example, the pack electronic processor **220** of the first battery pack **120A** detects a lack of power draw by the miter saw **110A** in response to the user releasing a trigger of the saw. In turn, the pack electronic processor **220** of the first battery pack **120A** sends a signal, via the pack transceiver, indicating that the miter saw **110A** has ceased being operated to the mobile device. In response to receiving the signal, the device electronic processor **310** of the mobile device **130** determines that the miter saw **110A** has ceased being operated.

Further, the device electronic processor **310** provides a second remote control command to the second battery pack **120B** in response to the determination that the miter saw **110A** has ceased being operated. For example, in response to determining that the miter saw **110A** has ceased being operated, the device electronic processor **310** transmits the second remote control command via the device transceiver **330**, and the second remote command is received by the second battery pack **120B**. The second remote control command is a request to turn the shop vacuum **110C** (i.e., the second power tool **110**) OFF (i.e., to deactivate a motor of the shop vacuum **110C**).

Further, the shop vacuum is controlled to turn OFF in response to the second remote control command received by the second battery pack **120B**. For example, the pack electronic processor **220** of the second battery pack **120B** relays the second remote control command to the tool electronic processor **200** of the shop vacuum **110C**. In response to the remote control command, the tool electronic processor **200** of the shop vacuum **110C** switches the shop vacuum **110** from the high power draw mode to the low power draw mode and deactivates the motor of the shop vacuum **110C**. Accordingly, the shop vacuum **110C** may be enabled and disabled essentially simultaneously with the miter saw **110A** without any user intervention. In other words, when the miter saw **110A** is activated by the user, the shop vacuum **110C** is activated, and when the miter saw **110A** is deactivated by the user, the shop vacuum **110C** is deactivated. This allows the dust collection process to be automated, which saves time for the user and provided a more efficient dust extraction.

9

In some embodiments, the first battery pack **120A** and the second battery pack **120B** communicate directly bypassing the mobile device **130**. The mobile device **130** may be used to communicatively connect the first battery pack **120A** and the second battery pack **120B**. The mobile device **130** may be used to pair (for example, Bluetooth® pairing) the first battery pack **120A** with the second battery pack **120B**. For example, a connection may be initiated using a graphical user interface (GUI) of a software application executing on the mobile device **130**. In this example, the mobile device **130** may detect that the battery packs **120A**, **120B** in wireless communication range; display an identifier for the battery packs **120A**, **120B** on the GUI; and allow a user to select on the GUI the first battery pack **120** and the second battery pack **120B** for pairing with each another. To pair the first battery pack **120A** with the second battery pack **120B**, the mobile device **130** may provide identification information, connection identification information, and/or password information for the connection to each of the first battery pack **120A** and the second battery pack **120B**. The first battery pack **120A** and the second battery pack **120B** use the identification information, connection information, and/or password information to subsequently establish a communication link or to communicate with each other. Particularly, the first battery pack **120A** and the second battery pack **120B** communicate directly to implement the method **600** as provided above.

Thus, embodiments described herein provide, among other things, a system and method for remote control of a power tool device.

What is claimed is:

1. A system for remote controlling a power tool device, the system comprising:
 - a tool electronic processor included in the power tool device, the power tool device including a battery pack interface; and
 - a battery pack configured to be coupled to the power tool device at the battery pack interface, the battery pack including:
 - a sensor configured to detect a type of the power tool device,
 - a pack transceiver, and
 - a pack electronic processor coupled to the pack transceiver and the sensor and in communication with the tool electronic processor, the pack electronic processor configured to:
 - determine that the power tool device is remotely controllable based on an output from the sensor,
 - receive, wirelessly via the pack transceiver, a remote control command from a mobile device, and
 - provide the remote control command to the power tool device,
 wherein the tool electronic processor is configured to control the power tool device to perform an action specified by the remote control command in response to receiving the remote control command from the pack electronic processor.
2. The system of claim 1, wherein the output from the sensor indicates a presence of a magnet that is proximate to the battery pack interface.
3. The system of claim 2, wherein the sensor is a Hall effect sensor.
4. The system of claim 1, wherein the output from the sensor indicates that a pole orientation representing that the power tool device is remotely controllable is present.

10

5. The system of claim 1, wherein the pack electronic processor is configured to determine that the power tool device is not remotely controllable based on a second output from the sensor.

6. The system of claim 5, wherein the second output from the sensor indicates that a magnet of the power tool device is in a first pole orientation.

7. The system of claim 1, further comprising:

- a second power tool device including a second tool electronic processor; and
- a second battery pack configured to be coupled to the second power tool device, the second battery pack including:
 - a second pack transceiver, and
 - a second pack electronic processor coupled to the second pack transceiver, the second pack electronic processor configured to:
 - determine that the coupled second power tool device is remotely controllable,
 - receive, wirelessly via the second pack transceiver, a second remote control command from the mobile device, and
 - provide the second remote control command to the second power tool device.

8. The system of claim 7, wherein the second pack electronic processor is configured to determine that the second power tool device is remotely controllable based on a second output from a second sensor within the second battery pack.

9. The system of claim 7, wherein the second pack electronic processor is further configured to:

- provide a command to the second power tool device in response to the pack electronic processor providing the remote control command to the power tool device.

10. A method for remote controlling a power tool device by a mobile device, the method comprising:

- determining, by a pack electronic processor of a battery pack, that the power tool device is remotely controllable tool type based on an output from a sensor of the battery pack,

receiving, wirelessly via a pack transceiver of the battery pack, a remote control command from the mobile device, and

- providing, via a communication link between the pack electronic processor and a tool electronic processor of the power tool device, the remote control command to the power tool device in response to determining that the power tool device is remotely controllable tool type,

wherein the tool electronic processor is configured to control the power tool device to perform an action specified by the remote control command in response to receiving the remote control command from the pack electronic processor.

11. The method of claim 10, wherein the output from the sensor indicates that a magnetic field is present proximate to a battery pack interface of the power tool device.

12. The method of claim 11, wherein the sensor senses a presence of a magnet provided proximate a battery pack interface of the power tool device.

13. The method of claim 10 further comprising:

- controlling, by the tool electronic processor, the power tool device in a low power mode prior to receiving the remote control command from the pack electronic processor.

11

14. The method of claim 13 further comprising:
switching, by the tool electronic processor, the power tool
device to be controlled in a high power mode to
perform the action in response to receiving the remote
control command.

15. A battery pack for remote controlling a power tool
device comprising:

an interface for coupling the battery pack to the power
tool device;

a sensor configured to detect a type of the power tool
device;

a pack transceiver; and

a pack electronic processor coupled to the pack trans-
ceiver and the sensor and in communication with a tool
electronic processor included in the power tool device,
the pack electronic processor configured to:

determine that the power tool device is remotely con-
trollable based on an output from the sensor,

receive, wirelessly via the pack transceiver, a remote
control command from a mobile device, and

provide the remote control command to the power tool
device,

12

wherein the tool electronic processor is configured to
control the power tool device in a high power mode in
response to receiving the remote control command.

16. The battery pack of claim 15, wherein the tool
electronic processor controls the power tool device in a low
power mode prior to receiving the remote control command.

17. The battery pack of claim 15, wherein the output from
the sensor indicates that a magnetic field is present.

18. The battery pack of claim 15, wherein the output from
the sensor indicates that a pole orientation of a magnet
proximate to a battery pack interface of the power tool
device is present.

19. The battery pack of claim 15, wherein the pack
electronic processor is configured to determine that the
power tool device is not remotely controllable based on a
second output from the sensor.

20. The battery pack of claim 19, wherein the second
output from the sensor indicates that a magnet of the power
tool device is in a first pole orientation.

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